Antonio Cruciani

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Short Bio

Antonio Cruciani is a postdoctoral researcher in Computer Science at Aalto University, working in Prof. Jukka Suomela's group. His research focuses on randomized algorithms for networks that can be observed only partially or that change continuously. One line designs distributed protocols that maintain global structure (expansion, connectivity, and distributed data structures) in peer-to-peer networks under adversarial churn. A second develops sampling algorithms for large static and temporal graphs whose sample complexity adapts to the structure of the input rather than to worst-case bounds. A third studies foundations of local computation. He obtained his Ph.D. in Computer Science from the Gran Sasso Science Institute in 2025, with a dissertation on centralized and distributed algorithms for evolving graphs. His work has appeared at ECML-PKDD, ICDM, DISC, PODC, SIROCCO, ICDCN, and SEA. He has held visiting research positions at IIT Madras, KTH Royal Institute of Technology, the University of Hamburg, and the University of Padua.

Fields of Research Expertise

Distributed Computing, Graph Mining, Randomized Algorithms, Temporal Graphs, Data Mining.

Current Position

Postdoctoral Researcher, Aalto University, Computer Science. Research Group of Prof. Jukka Suomela. — *Espoo, FI*, *Mar. 2025 – Present*

Education

Ph.D. in Computer Science, Gran Sasso Science Institute (GSSI). — *L'Aquila, IT*, *Mar. 2025*
Thesis: *Models and Algorithms for Temporal Betweenness Centrality and Dynamic Distributed Data Structures.* **Grade:** Excellent.
Supervisors: Francesco Pasquale (University of Rome "Tor Vergata"), Pierluigi Crescenzi (GSSI)
Reviewers: Christian Scheideler (UPB), Fabio Vandin (UNIPD).

M.S. in Computer Science, University of Rome "Tor Vergata", — *Rome, IT*, *2020*
Thesis: *Dynamic Random Graphs and unstructured P2P networks, analysis of two models inspired by the Bitcoin network.* Summa cum Laude.
Supervisor: Francesco Pasquale (University of Rome "Tor Vergata")

B.S. in Computer Science, University of Rome "Tor Vergata". — *Rome, IT*, *2017*
Thesis: *Efficient learning methods for playlist prediction.*
Supervisor: Giorgio Gambosi (University of Rome "Tor Vergata")

Visiting Research Periods

IIT Madras, Department of Computer Science and Engineering. Two research stays (7 and 3 months). — *Chennai, IN*, *Aug. 2023 – Feb. 2024*, *Aug. – Oct. 2024*
Host Professor: John Augustine

University of Padua, Department of Computer Science and Engineering. — *Padova, IT*, *July 2024, Feb. 2026*
Host Professor: Leonardo Pellegrina

KTH Royal Institute of Technology. *Stockholm, SE*
Host Professor: Aristides Gionis *June 2025*

University of Hamburg, Department of Computer Science. *Hamburg, DE*
Host Researcher: Thorsten Götte *January 2025*

Fondazione Ugo Bordoni, Big Data Analytics Lab. *Rome, IT*
Host Researcher: Giambattista Amati *Feb. – Nov. 2020*

Teaching Experience

Invited Teacher, *CS-E4565 - Combinatorics of Computation D.* (MSc), Aalto University (lecture delivery, guided problem solving, office hours, grading). **Topics:** probabilistic method, theory of concentration, expander graphs. *Finland, FI*
April - May. 2026

Invited Teacher, *CS-E4565 - Combinatorics of Computation D.* (MSc), Aalto University (lecture delivery, guided problem solving, office hours, grading). **Topics:** introduction to probability, the probabilistic method, and derandomization across four sessions. *Finland, FI*
April - May. 2025

Teaching Assistant, Computability and Computational Complexity Theory (BSc), University of Rome "Tor Vergata". (lecture delivery and guided problem solving) **Topics:** computability and computational complexity. *Rome, IT*
Oct. 2018 – Jun. 2019

Teaching Assistant, Computer Programming in C (BSc), University of Rome "Tor Vergata". Led exercise sessions focused on problem solving and programming practice, supported students during in-class exercises, and graded weekly homework assignments. *Rome, IT*
Oct. 2017 – Jun. 2018

Languages

- English (fluent)
- Italian (mother tongue)

Publications

In case of theoretical computer science conferences (⧫), authors are sorted alphabetically, otherwise (★) by contribution.

Selected Publications

1. MANTRA: Temporal Betweenness Centrality Approximation Through Sampling. **ECML-PKDD 2024** (★). *Single-authored.*
2. Fast Percolation Centrality Approximation with Importance Sampling. **ICDM 2025** (★).
3. Highly Dynamic and Fully Distributed Data Structures. **DISC 2025** (⧫) (brief announcement; full version under submission).
4. Maintaining a Bounded Degree Expander in Dynamic Peer-to-Peer Networks. **DISC 2025; SIROCCO 2026** (⧫). *Single-authored.*
5. It Does Not Matter How You Define Locally Checkable Labelings. **PODC 2026**; **DISC 2026** (⧫).
6. PROPAGATE: A Seed Propagation Framework to Compute Distance-Based Metrics on Very Large Graphs. **ECML-PKDD 2023** (★).

Conference Proceedings

[1] M. T. E. Abedesselam, **A. Cruciani**, F. Giacomelli, L. Kiffer, and F. Pasquale. "How Bitcoin Forms Its Network: Peer-Table Sampling and Structural Properties". In: *(TO APPEAR) Stabilization, Safety, and Security of Distributed Systems - International Symposium, SSS 2026* (♦), *Gothenburg, Sweden, October 9-11, 2026, Proceedings*. Lecture Notes in Computer Science. Springer, 2026.

[2] **A. Cruciani**, A. Das, A. Raevskaya, and J. Suomela. "It Does Not Matter How You Define Locally Checkable Labelings". In: *(TO APPEAR) 40th International Symposium on Distributed Computing (DISC 2026)* (♦). Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2026. Acceptance rate = 26%.
Brief announcement: *Proceedings of the ACM Symposium on Principles of Distributed Computing*, PODC '26 (♦), acceptance rate = 22%, doi:10.1145/3796701.3815923.

[3] **A. Cruciani**, A. Das, M. Equi, H. Lievonen, D. Luong-Le, A. Modanese, and J. Suomela. "Is a LOCAL Algorithm Computable?" In: *(TO APPEAR) 40th International Symposium on Distributed Computing (DISC 2026)* (♦). Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2026. Acceptance rate = 26%.
Brief announcement: *Proceedings of the ACM Symposium on Principles of Distributed Computing*, PODC '26 (♦), acceptance rate = 22%, doi:10.1145/3796701.3815908.

[4] **A. Cruciani**. "Maintaining a Bounded Degree Expander in Dynamic Peer-to-Peer Networks". In: *Structural Information and Communication Complexity - 33rd International Colloquium, SIROCCO 2026* (♦), *Durham, UK, June 9-11, 2026, Proceedings*. Lecture Notes in Computer Science. Springer, 2026. DOI: 10.1007/978-3-032-26465-7_11. URL: https://doi.org/10.1007/978-3-032-26465-7_11. Acceptance rate = 43%.
Brief announcement: *39th International Symposium on Distributed Computing (DISC 2025)* (♦), 53:1–53:7, acceptance rate = 30.5%, doi:10.4230/LIPIcs.DISC.2025.53.

[5] **A. Cruciani** and L. Pellegrina. "Fast Percolation Centrality Approximation with Importance Sampling". In: *IEEE International Conference on Data Mining, ICDM 2025* (★), *Washington DC, USA, November 12-15, 2025*. IEEE, 2025, pp. 1125–1134. DOI: 10.1109/ICDM65498.2025.00121. URL: https://doi.org/10.1109/ICDM65498.2025.00121. Acceptance rate = 8.9%.

[6] J. Augustine, **A. Cruciani**, and I. A. Gillani. "Brief Announcement: Highly Dynamic and Fully Distributed Data Structures". In: *39th International Symposium on Distributed Computing (DISC 2025)* (♦). Ed. by D. R. Kowalski. Vol. 356. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025, 47:1–47:7. ISBN: 978-3-95977-402-4. DOI: 10.4230/LIPIcs.DISC.2025.47. URL: https://drops.Dagstuhl.de/entities/document/10.4230/LIPIcs.DISC.2025.47. Acceptance rate = 30.5%.

[7] A. Balliu, C. Coupette, **A. Cruciani**, F. d'Amore, M. Equi, H. Lievonen, A. Modanese, D. Olivetti, and J. Suomela. "New Limits on Distributed Quantum Advantage: Dequantizing Linear Programs". In: *39th International Symposium on Distributed Computing (DISC 2025)* (♦). Ed. by D. R. Kowalski. Vol. 356. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025, 11:1–11:22. ISBN: 978-3-95977-402-4. DOI: 10.4230/LIPIcs.DISC.2025.11. URL: https://drops.Dagstuhl.de/entities/document/10.4230/LIPIcs.DISC.2025.11. Acceptance rate = 30.5%.

[8] **A. Cruciani**. "MANTRA: Temporal Betweenness Centrality Approximation Through Sampling". In: *Machine Learning and Knowledge Discovery in Databases. Research Track - European Conference, ECML PKDD 2024* (★), *Vilnius, Lithuania, September 9-13, 2024, Proceedings, Part I*. Ed. by A. Bifet, J. Davis, T. Krilavicius, M. Kull, E. Ntoutsi, and I. Zliobaite. Vol. 14941. Lecture Notes in Computer Science. Springer, 2024, pp. 125–143. DOI: 10.1007/978-3-031-70341-6_8. URL: https://doi.org/10.1007/978-3-031-70341-6_8. Acceptance rate = 24%.

[9] G. Amati, **A. Cruciani**, D. Pasquini, P. Vocca, and S. Angelini. "propagate: A Seed Propagation Framework to Compute Distance-Based Metrics on Very Large Graphs". In: *Machine Learning and Knowledge Discovery in Databases: Research Track - European Conference, ECML PKDD 2023* (★), *Turin, Italy, September 18-22, 2023, Proceedings, Part III*. Ed. by D. Koutra, C. Plant, M. G. Rodriguez, E. Baralis, and F. Bonchi. Vol. 14171. Lecture Notes in Computer Science. Springer, 2023, pp. 671–688. DOI: 10.1007/978-3-031-43418-1_40. URL: https://doi.org/10.1007/978-3-031-43418-1_40. Acceptance rate = 24%.

[10] R. Becker, P. Crescenzi, **A. Cruciani**, and B. Kodric. "Proxying Betweenness Centrality Rankings in Temporal Networks". In: *21st International Symposium on Experimental Algorithms, SEA 2023* (♦)*, July 24-26, 2023, Barcelona, Spain*. Ed. by L. Georgiadis. Vol. 265. LIPIcs. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2023, 6:1–6:22. DOI: 10.4230/LIPICS.SEA.2023.6. URL: https://doi.org/10.4230/LIPIcs.SEA.2023.6. Acceptance rate = 50%.

[11] **A. Cruciani** and F. Pasquale. "Dynamic graph models inspired by the Bitcoin network-formation process". In: *24th International Conference on Distributed Computing and Networking, ICDCN 2023* (♦)*, Kharagpur, India, January 4-7, 2023*. ACM, 2023, pp. 125–134. DOI: 10.1145/3571306.3571398. URL: https://doi.org/10.1145/3571306.3571398. Acceptance rate = 34%.

[12] **A. Cruciani** and F. Pasquale. "Brief Announcement: Dynamic Graph Models for the Bitcoin P2P Network: Simulation Analysis for Expansion and Flooding Time". In: *Stabilization, Safety, and Security of Distributed Systems - 24th International Symposium, SSS 2022* (♦)*, Clermont-Ferrand, France, November 15-17, 2022, Proceedings*. Ed. by S. Devismes, F. Petit, K. Altisen, G. A. D. Luna, and A. F. Anta. Vol. 13751. Lecture Notes in Computer Science. Springer, 2022, pp. 335–340. DOI: 10.1007/978-3-031-21017-4_23. URL: https://doi.org/10.1007/978-3-031-21017-4_23.

[13] G. Amati, S. Angelini, **A. Cruciani**, G. Fusco, G. Gaudino, D. Pasquini, and P. Vocca. "Topic Modeling by Community Detection Algorithms". In: *OASIS@HT 2021* (★)*: Proceedings of the 2021 Workshop on Open Challenges in Online Social Networks, Virtual Event, Ireland, 30 August 2021*. Ed. by B. Guidi, A. Michienzi, and L. Ricci. ACM, 2021, pp. 15–20. DOI: 10.1145/3472720.3483622. URL: https://doi.org/10.1145/3472720.3483622.

[14] **A. Cruciani**, D. Pasquini, G. Amati, and P. Vocca. "About Graph Index Compression Techniques". In: *Proceedings of the 10th Italian Information Retrieval Workshop* (★)*, Padova, Italy, September 16-18, 2019*. Ed. by M. Agosti, E. D. Buccio, M. Melucci, S. Mizzaro, G. Pasi, and F. Silvestri. Vol. 2441. CEUR Workshop Proceedings. CEUR-WS.org, 2019, pp. 21–24. URL: https://ceur-ws.org/Vol-2441/paper23.pdf.

Preprints

[15] J. Augustine, **A. Cruciani**, and I. A. Gillani. *Highly Dynamic and Fully Distributed Data Structures*. 2025. arXiv: 2409.10235 [cs.DC]. URL: https://arxiv.org/abs/2409.10235.

[16] T. E. M. Abedesselam, **A. Cruciani**, F. Giacomelli, L. Kiffer, and F. Pasquale. *How Bitcoin Forms Its Network: Peer-Table Sampling and Structural Properties*. 2026. arXiv: 2609.07609 [cs.NI]. URL: https://arxiv.org/abs/2609.07609.

[17] J. Augustine, **A. Cruciani**, M. Kumar, S. Kutten, and D. Peleg. *Gossip in an Anonymous Capacitated Networks*. 2026. (Working Paper).

PhD Thesis

[18] **A. Cruciani**. *Models and Algorithms for Temporal Betweenness Centrality and Dynamic Distributed Data Structures*. 2025. URL: https://iris.gssi.it/handle/20.500.12571/34764.

Invited Talks & Seminars

- **2024:**
  - Approximating Distance-based metrics through sampling (IIT Madras, Chennai, IN);
  - Maintaining Distributed Data Structures in Dynamic Peer-to-Peer Networks (IIT Madras, Chennai, IN);
  - On the Temporal Betweenness Centrality (IIT Madras, Chennai, IN);
  - Maintaining Distributed Data Structures in Dynamic Peer-to-Peer Networks (Aalto University, Espoo, FI; online);
  - Computing Distance-based metrics on Very Large Graphs (University of Padua, Padova, IT).
- **2019:**
  - Iterative Compression technique for NP-Hard problems on Graphs (University of Rome "Tor Vergata", Rome, IT).

Conference & Workshop Presentations

- **2026:**
  - Maintaining a Bounded Degree Expander in Dynamic Peer-to-Peer Networks (SIROCCO, Durham, UK);
- **2025:**
  - Fast Percolation Centrality Approximation with Importance Sampling (ICDM, Washington DC, USA);
  - Maintaining a Bounded Degree Expander in Dynamic Peer-To-Peer Networks (DISC, Berlin, DE);
  - Maintaining Distributed Data Structures in Dynamic Peer-to-Peer Networks (DISC, Berlin, DE);
  - Maintaining Distributed Data Structures in Dynamic Peer-to-Peer Networks (Joint Estonian-Latvian Theory Days, Riga, LV);
  - Fast Percolation Centrality Approximation with Importance Sampling (Helsinki Algorithms & Theory Days, Helsinki, FI).
- **2024:**
  - MANTRA: Temporal Betweenness Centrality Approximation through Sampling (ECML-PKDD, Vilnius, LT).
- **2023:**
  - PROPAGATE: A Seed Propagation Framework to Compute Distance-Based Metrics on Very Large Graphs (ECML-PKDD, Turin, IT);
  - Proxying Betweenness Centrality Rankings in Temporal Networks (SEA, Barcelona, ES);
  - Dynamic graph models inspired by the Bitcoin network-formation process (ICDCN, Kharagpur, IN).
- **2022:**
  - Dynamic graph models for the Bitcoin P2P network: simulation analysis for expansion and flooding time (SSS, Clermont-Ferrand, FR).
- **2019:**
  - About Graph Index Compression Techniques (IIR, Padova, IT).

Additional Training

Bertinoro International Spring School	*Bertinoro, IT* *Mar. 2022*
European Summer School on Learning in Games, Markets, and Online Decision Making	*Rome, IT* *Sep. 2021*
Max Planck Advanced Course on the Foundations of Computer Science (Convex Optimization) (online)	*Saarbrucken, DE* *Jul. - Aug. 2021*
Algorithmic Tools for Massive Network Analytics (online)	*Pisa, IT* *May - Jun. 2021*
Max Planck Advanced Course on the Foundations of Computer Science (Market Design and Computational Fair Division) (online)	*Saarbrucken, GE* *August 2020*
Algorithms and computational models for large-scale data analysis. University of Rome: "La Sapienza". By Silvio Lattanzi (Google Research).	*Rome, IT* *August 2019*

Academic Service

Program Committee

PC-Member: 28th International Conference on Distributed Computing and Networking (ICDCN 2027).
PC-Member: The 30th Conference on Principles of Distributed Systems (OPODIS 2026).
PC-Member: The 28th International Symposium on Stabilization, Safety, and Security of Distributed Systems (SSS 2026).

Reviewing

Conferences
Reviewer ICALP-PODC-DISC 2026
Reviewer STACS-PODC-WSDM-SDM-SEA-ISAAC-FSTTCS-ICANN 2025
Reviewer FUN 2020-2024
Reviewer AAMAS 2023

Journals
Reviewer: The Review of Socionetwork Strategies

Organization

Organizer of the workshop Helsinki Theory Days 2026

Students Supervision

Bachelor's Students: Giacomo Rivetti (2025, co-supervised with Francesco Pasquale).
Master's Students: Andrea Bonifati (2026, co-supervised with Francesco Pasquale).

Software Packages

- PROPAGATE (Java), an efficient algorithm for approximating various distance-based metrics (i.e., average distance, effective diameter, diameter and connectivity rate).
- DREG (Python), a dynamic expander graph generator.
- TSBPROXY (Julia), a suite of efficient proxies for the temporal betweenness centrality rankings.
- MANTRA (Julia), an efficient framework for approximating the temporal betweenness centrality using sampling.
- FEPIC (Julia), an efficient approximation algorithm for the (doubly normalized) percolation centrality.
- PERCIS (C++), a fast approximation algorithm for the percolation centrality that uses importance sampling.

Contact References

- Jukka Suomela, Aalto University: jukka.suomela@aalto.fi
- Francesco Pasquale, University of Rome Tor Vergata: pasquale@mat.uniroma2.it
- John Augustine, IIT Madras: augustine@cse.iitm.ac.in
- Leonardo Pellegrina, University of Padua: leonardo.pellegrina@unipd.it